

proxy server), and Puppet (a configuration management tool written in Ruby and released under the General Public License or GPL).

Interestingly, Microsoft has also been captioning open source strengths amongst the benefits of Azure, its cloud-computing platform. The Windows Azure software development kit for PHP and the Windows Azure Tools for Eclipse make it easy for programmers to deploy their PHP applications to the Azure cloud, not to forget the support for command-line developers to leverage scripting skills in the deployment of existing PHP applications. Then there is the Windows Azure Companion, which apparently eases the task of deploying open source community applications such as WordPress, SugarCRM, Drupal, etc, onto Windows Azure. Microsoft has also been working with many open source developers to put more and more FOSS programs on Azure.

The focus on open standards

It is clear that there are tons of open source software for cloud computing; you just need to make sure you pick the right ones. Remember, it is not just about open source software; you need to ensure that your choice is based on open standards too. That is precisely what the OpenStack project and consortium are all about.

OpenStack is a collection of open source technologies that deliver a highly-scalable cloud operating system, based completely on open standards. OpenStack has two interrelated projects: OpenStack Compute and OpenStack Object Storage. Compute is for provisioning and managing large groups of virtual private servers, while Object Storage is for creating redundant, scalable object storage using clusters of commodity servers. Object Storage can handle even petabytes of data! OpenStack integrates code from NASA's Nebula platform as well as Rackspace's Cloud Files platform, and is released under the Apache 2.0 licence. OpenStack's forte is large-scale computational prowess, like those needed for DNA modelling, space research, and the likes.

More than anything else, OpenStack.org is a great example of how industry players are converging to set open standards in place to build clouds. OpenStack is backed by big names such as Rackspace, NASA, Dell, Citrix, Cisco and Canonical, not to forget the large global open source community. The team feels that an open development model is the only way to foster badly-needed cloud standards, remove the fear of proprietary lock-in for cloud customers, and create a large ecosystem that spans cloud providers.

"It is important for customers to have choices and not be tied down to a proprietary platform. Open standards for the cloud will play a key role in enhancing cloud adoption. When I say open standards, I don't necessarily

imply open source, but the fact that customers will have choices and the ability to stitch together solutions that meet their needs. Open standards and specifications would ensure that different products and tools can co-exist to deliver the cloud. Open source will be a key enabler to providing that choice. Such initiatives are already under way—one of them being the OpenStack consortium, in which Dell is a key participant. We believe such initiatives, and the existing open source ecosystem, will evolve cloud computing to the next stage," says Sinha.

Open source, the best for India

There are loads of open source solutions for cloud computing out there. New ones, old ones, modified or extended ones, et al. Cloud computing being a nascent space, there is still lots more for the community to do. Sinha says, "Cloud computing throws up new problems to tackle, and new possibilities to address through the open source community. While some of the existing OSS contributions would be a great natural extension to the cloud world, there are several areas in the cloud-computing puzzle that require newer open source offerings."

Of course, open source software and the community will continue to do its bit—and more—for cloud computing. "The ongoing impact of open source, the whole concept of SaaS meeting the cloud, is going to be a major trend in India. The open source approach in cloud computing will definitely grow in the future. Different organisations with different requirements can customise the applications according to their own use. New and advanced components, which are more flexible, transparent and cost-effective, can be integrated with the applications at any point of time, enhancing the capabilities of the system," says Paul.

Advani sums up: "The availability and cost benefit of open source allows India to build cloud solutions for India. Service providers can adopt Ubuntu Enterprise Cloud, for instance, and provide a local, world-class scalable cloud provision for local businesses. The risk of a cloud in the proprietary world is reliance on a single vendor, external to the country, with service-level agreements that are liable to change. Widespread adoption of an open cloud solution is a much better route to avoid this risk. Of course, the open source cloud has all the advantages: it's cost-effective, there's no vendor lock-in, and you have access to the code, so you are not worried about getting stuck with the wrong technology!" That, after all, is very important for nascent technologies like the cloud. 

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